Lab 7: Fuzzing for Input Validation Bugs, Screenshots

Step 2: Running the buggy application

```
vasut@Vasu MINGW64 ~/Documents/SE LAB 6/L6_L7_39_password_policy/lab6 (main)
$ cd ../lab7
python password_policy.py
==== Running first_character_type ====
Enter a password:
Traceback (most recent call last):
  File "C:\Users\vasut\Documents\SE LAB 6\L6_L7_39_password_policy\lab7\password_policy.py", line 33, in <module>
    main()
  File "C:\Users\vasut\Documents\SE LAB 6\L6_L7_39_password_policy\lab7\password_policy.py", line 19, in main
    result = first_character_type(user_input)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\vasut\Documents\SE LAB 6\L6_L7_39_password_policy\lab7\password_policy.py", line 2, in first_character_type
    ch = pw[0]
         ~~^^^
IndexError: string index out of range
```

Figure 1: Original program crashes with IndexError when an empty password is entered.

Step 4: Fuzz tests on the buggy code

```
_____ test_digit_ratio_in_range _____

    @given(st.text())
>   def test_digit_ratio_in_range(s):
                ^^^

test_password_policy.py:47:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
test_password_policy.py:48: in test_digit_ratio_in_range
    r = digit_ratio(s)
        ^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

pw = ''

    def digit_ratio(pw):
>       return sum(c.isdigit() for c in pw) / len(pw)
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       ZeroDivisionError: division by zero
E       Failing test case: test_digit_ratio_in_range(
E           s='',
E       )

password_policy.py:14: ZeroDivisionError
=============================== warnings summary ===============================
..\..\..\..\..\..\Program Files\Python311\Lib\site-packages\anyio\pytest_plugin.py:14
  C:\Program Files\Python311\Lib\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
=========================== short test summary info ============================
FAILED test_password_policy.py::test_first_character_type_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_password_policy.py::test_split_credentials_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_password_policy.py::test_digit_ratio_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_password_policy.py::test_digit_ratio_in_range - ZeroDivisionError: division by zero
========== 4 failed, 2 passed, 1 warning in 2.85s ==========
```

Figure 2: Hypothesis finds crashes in all three functions (empty string and None inputs): 4 failed, 2 passed.

Step 5: Running the fixed application

```
vasut@Vasu MINGW64 ~/Documents/SE LAB 6/L6_L7_39_password_policy/lab7 (main)
$ python password_policy.py
==== Running first_character_type ====
Enter a password:
First Character Type: empty

==== Running split_credentials ====
Enter user:password (e.g. alice:Secret1): abc
Invalid format - expected user:password

==== Running digit_ratio ====
Enter a password to measure:
Digit Ratio: 0.0
```

Figure 3: Fixed program handles an empty password, a missing colon and empty input without crashing.

Step 6: Fuzz tests on the fixed code

```
vasut@Vasu MINGW64 ~/Documents/SE LAB 6/L6_L7_39_password_policy/lab7 (main)
$     python -m pytest test_password_policy.py
============================================================ test session starts ============================================================
platform win32 -- Python 3.11.0, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\vasut\Documents\SE LAB 6\L6_L7_39_password_policy\lab7
plugins: hypothesis-6.170.15, platformdirs-4.12.2, anyio-4.14.2, typeguard-4.6.0
collected 6 items

test_password_policy.py ......                                                                                                      [100%]

============================================================ warnings summary ============================================================
..\..\..\..\..\..\Program Files\Python311\Lib\site-packages\anyio\pytest_plugin.py:14
  C:\Program Files\Python311\Lib\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
========================================================== 6 passed, 1 warning in 2.59s ==========================================================

vasut@Vasu MINGW64 ~/Documents/SE LAB 6/L6_L7_39_password_policy/lab7 (main)
$
```

Figure 4: All fuzz tests pass after hardening the functions: 6 passed.